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UNIVERSITY INSTITUTE OF COMPUTING

PROJECT REPORT ON
LifeOS – Personal Digital Command Center

OBJECT ORIENTED PROGRAMMING
25CAH-201

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1. ABSTRACT

LifeOS – Personal Digital Command Center is a desktop-based productivity management application developed using C++, Qt 6, and SQLite.

The purpose of this project is to provide users with a centralized platform to manage their daily activities, organize tasks, maintain notes, track habits, schedule events, and monitor personal finances.

In modern life, individuals often depend on multiple applications to manage their responsibilities. This can make information difficult to organize and increase the chances of missing important deadlines and activities.

LifeOS addresses this problem by integrating different productivity-related functionalities into one desktop application. It provides an interactive dashboard that helps users monitor their activities and review their productivity.

The application uses SQLite for local database management and Qt 6 Widgets for its graphical user interface. C++ handles the application's functionality and database operations.

The project demonstrates the practical implementation of programming concepts, database connectivity, graphical user interface development, and software engineering principles.

2. INTRODUCTION

LifeOS – Personal Digital Command Center is a desktop application developed using C++ and Qt. The main purpose of this project is to provide users with a single platform where they can manage their daily activities, tasks, finances, notes, habits, and schedules.

In daily life, people often use different applications to manage their work. LifeOS aims to bring these important activities together in one organized application.

The application provides a user-friendly graphical interface and uses an SQLite database to store information.

3. OBJECTIVES OF THE PROJECT

The main objectives of LifeOS are:

1. To develop an all-in-one personal management application.
2. To help users organize and manage their daily tasks.
3. To provide a system for tracking income and expenses.
4. To help users maintain daily habits and routines.
5. To provide a place for storing personal notes.
6. To organize events and important dates.
7. To display useful information through a dashboard.
8. To understand practical implementation of C++, Qt, and database connectivity.

4. PROBLEM STATEMENT

Managing daily activities through different applications can be time-consuming and confusing. Users may need separate applications for task management, financial tracking, notes, and habit tracking.

The objective of LifeOS is to reduce this difficulty by providing multiple personal management features within one desktop application.

5. TECHNOLOGIES USED

The following technologies and tools are used in the development of LifeOS:

Technology	Purpose
C++	Main programming language
Qt 6	Graphical User Interface development
Qt Widgets	Creating application windows and controls
SQLite	Storing application data
Qt SQL	Connecting the application with the database
CMake	Project configuration and build management
MSYS2 UCRT64	Development environment

6. SYSTEM REQUIREMENTS

Hardware Requirements

- Computer or laptop
- Minimum 4 GB RAM
- Keyboard and mouse
- Sufficient storage space

Software Requirements

- Windows operating system

- C++ compiler
- Qt 6
- MSYS2 UCRT64
- CMake
- SQLite

7. OVERVIEW OF THE APPLICATION

LifeOS contains different modules that help users manage their personal activities.

The main modules are:

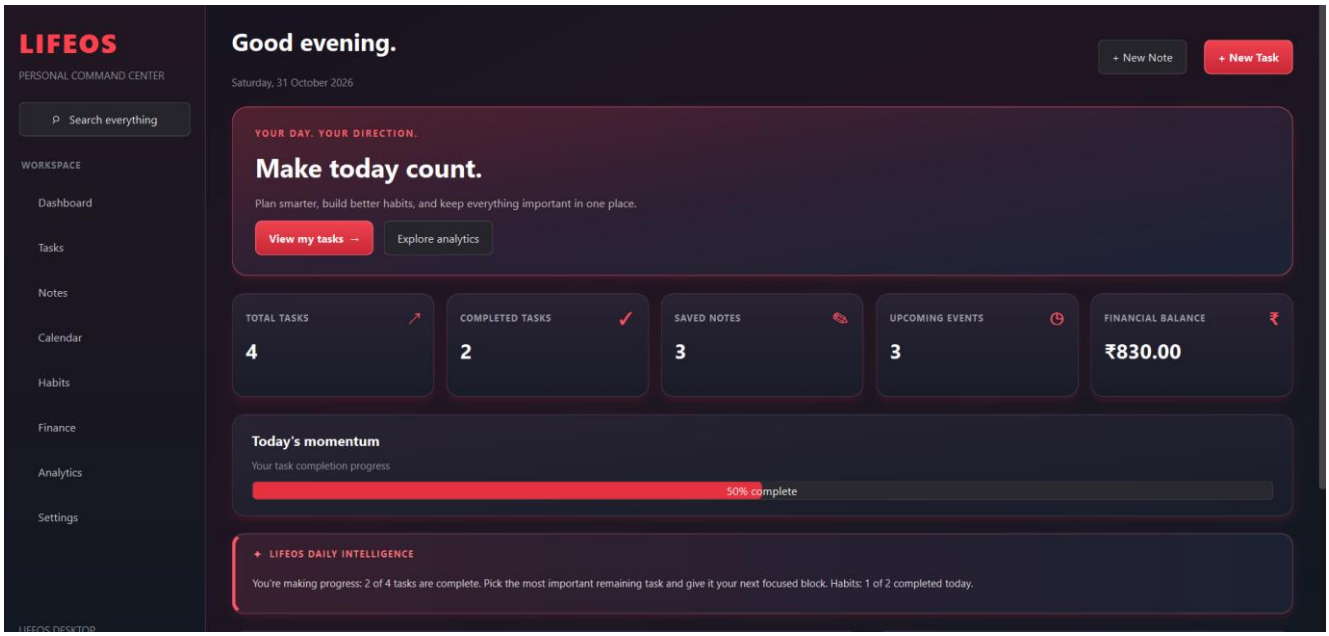
1. Dashboard
2. Task Manager
3. Calendar
4. Habit Tracker
5. Personal Finance
6. Notes
7. Analytics
8. Settings

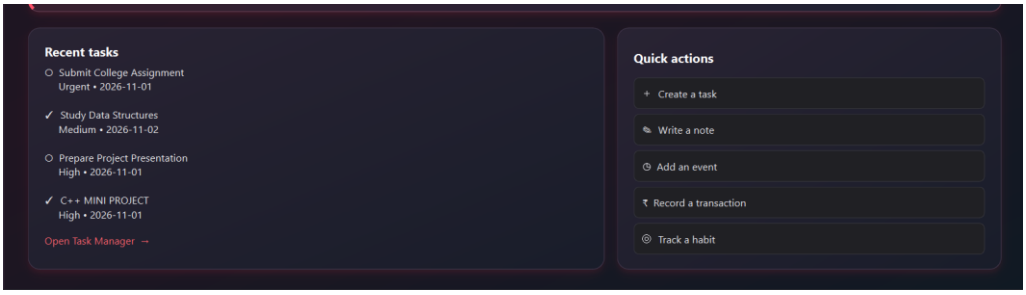
Each module has a separate purpose, but all are accessible through the main application interface.

8. DASHBOARD MODULE

The Dashboard is the main screen of LifeOS. It provides an overview of important information and allows users to navigate to different modules.

It displays summary information such as completed tasks, daily habits, monthly expenses, and productivity.





The dashboard helps users access different features quickly and view important information in one place.

9. TASK MANAGER MODULE

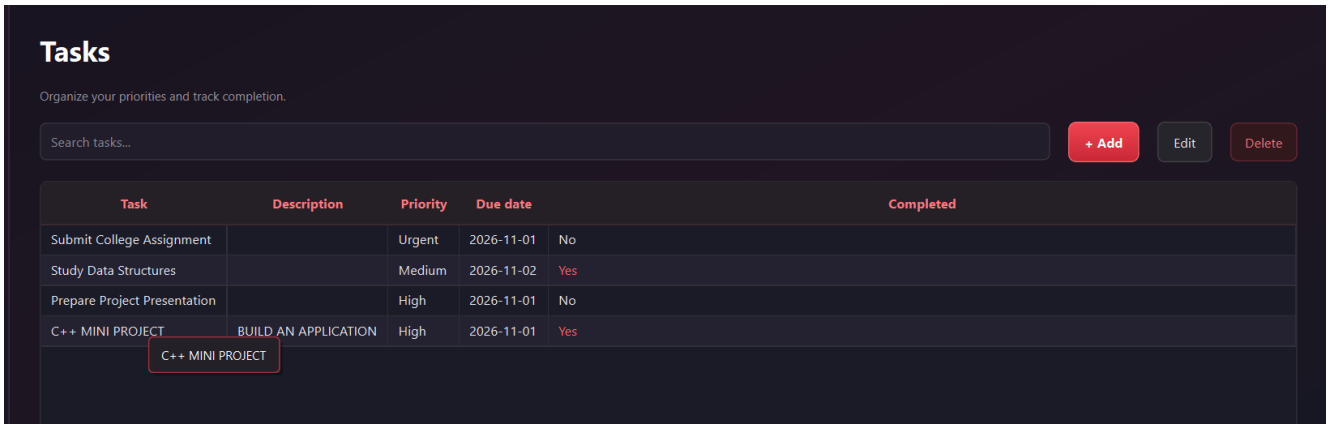
The Task Manager helps users create, edit, delete, search, and manage their daily tasks.

Users can enter task details such as:

- Task title
- Description
- Priority
- Due date

- Completion status

Users can also mark tasks as completed and view their task list.

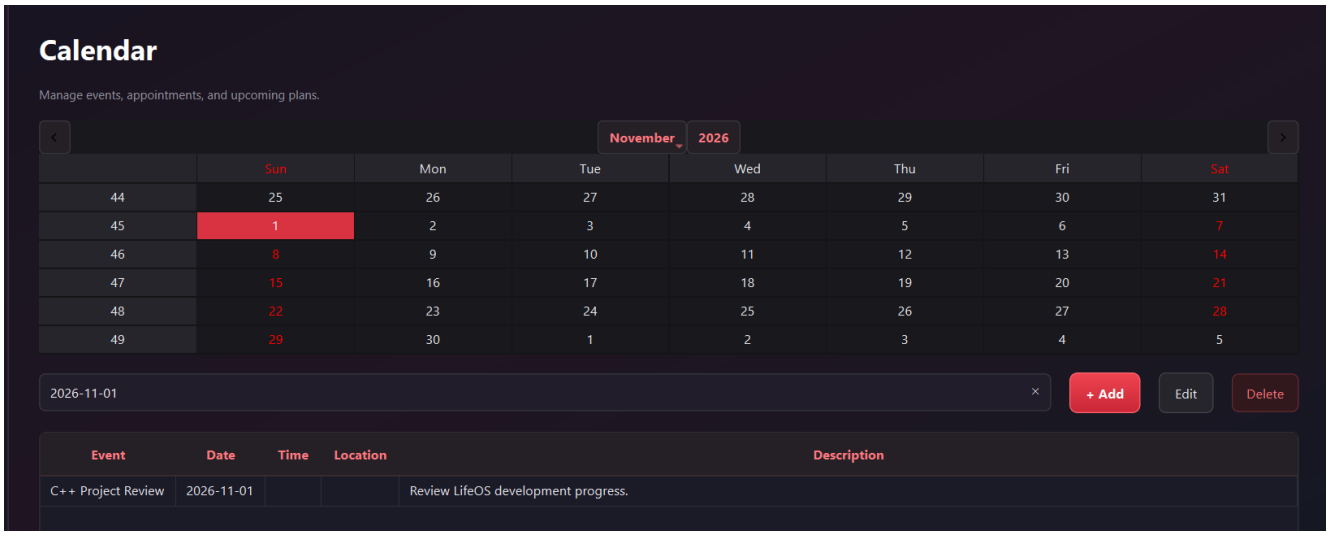


This module helps users organize their work and keep track of pending activities.

10. CALENDAR MODULE

The Calendar module is designed to help users organize important dates, events, and deadlines.

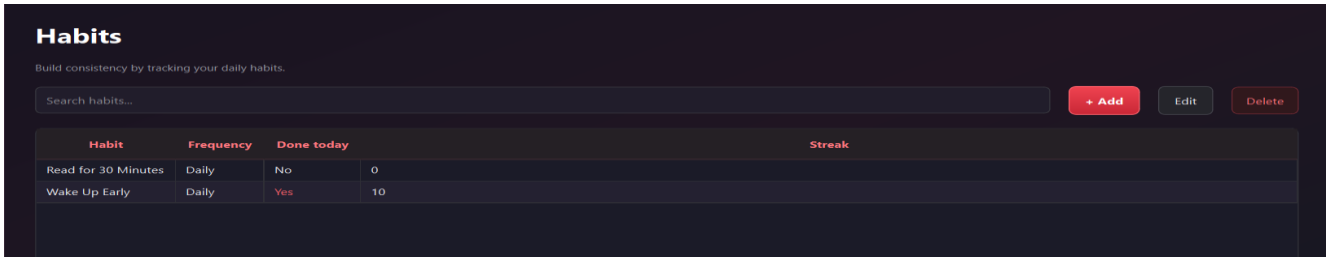
It provides a dedicated place for managing schedules and planning future activities.



11. HABIT TRACKER MODULE

The Habit Tracker is designed to help users build and maintain positive daily routines.

Users can use this module to monitor their habits and maintain consistency in their daily activities.



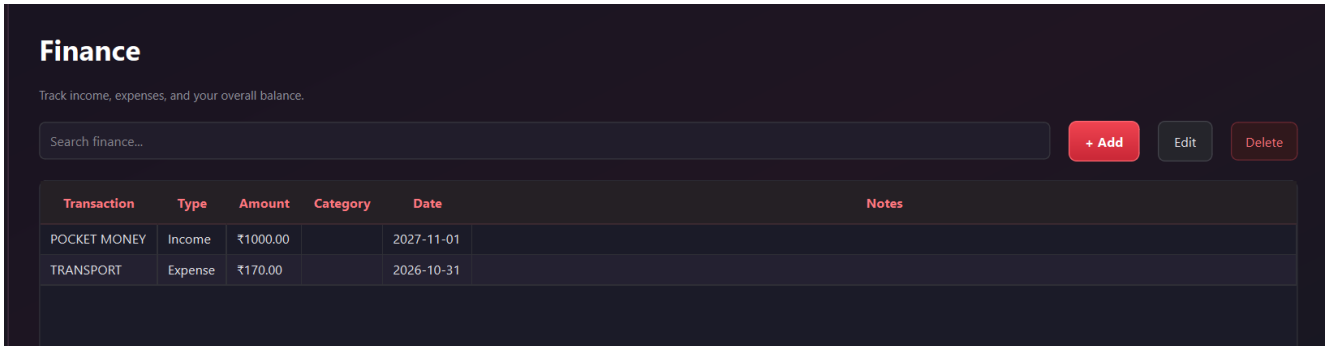
12. PERSONAL FINANCE MODULE

The Personal Finance module helps users manage their income and expenses.

It is designed to provide a simple way to record financial transactions and monitor spending.

The module includes features such as:

- Adding income
- Recording expenses
- Viewing transaction records
- Tracking financial information

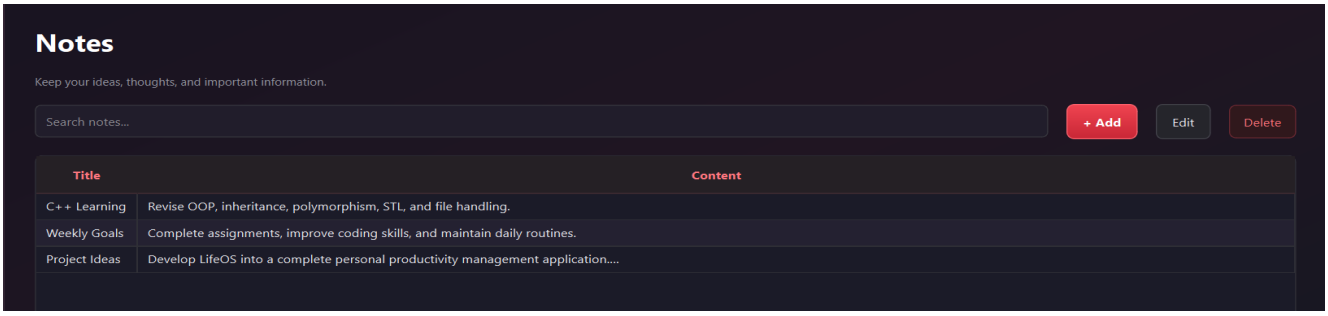


This module helps users maintain a record of their financial activities.

13. NOTES MODULE

The Notes module provides a space for writing and storing important information, ideas, and reminders.

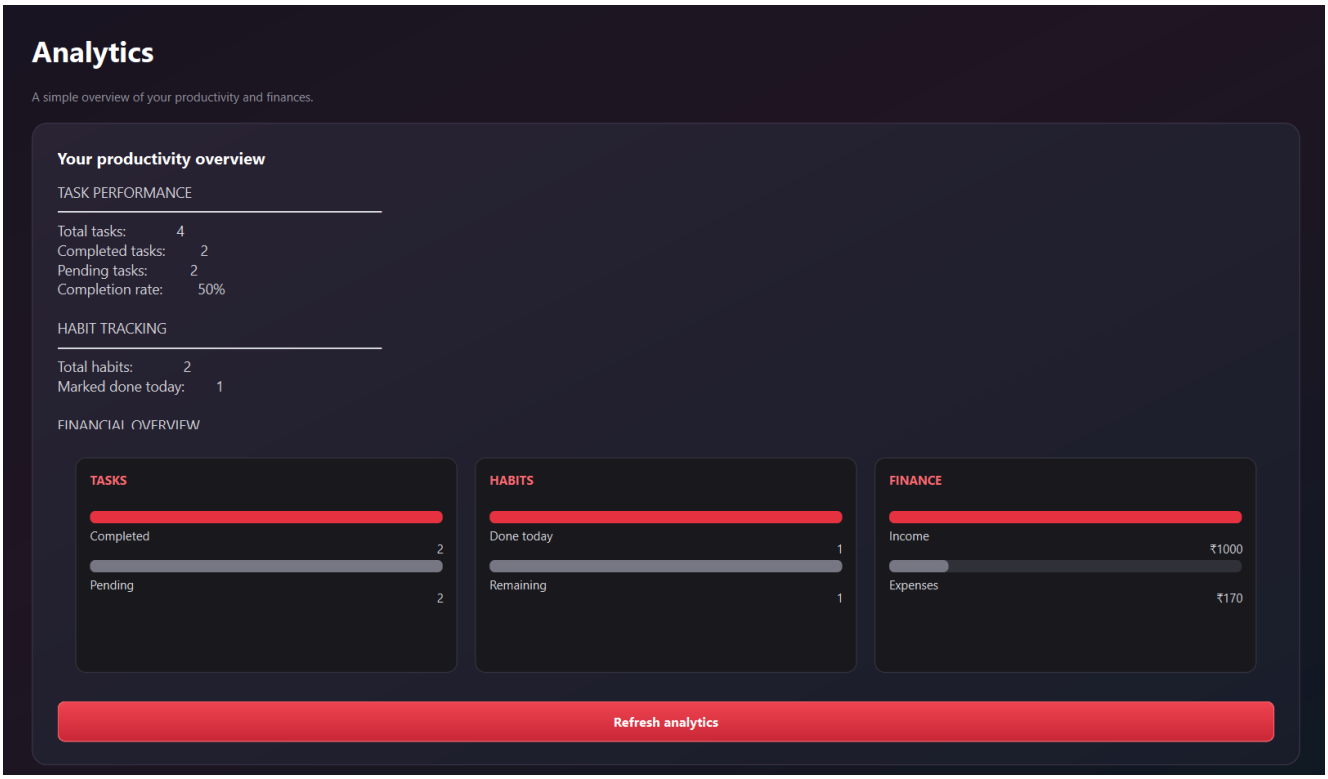
It helps users keep their personal information organized within the application.



14. ANALYTICS MODULE

The Analytics module is designed to present useful information about user activities.

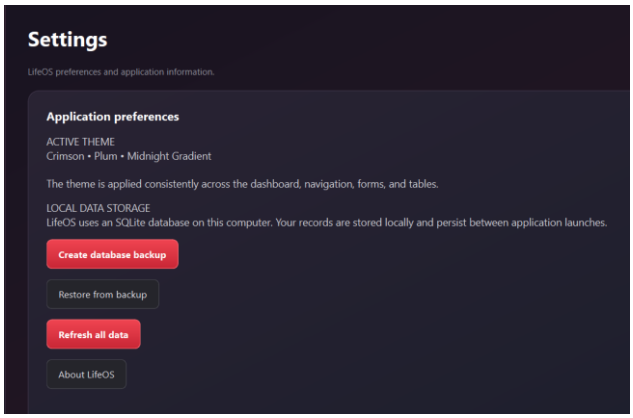
It can help users understand their productivity, task completion, habits, and financial activity through summarized information.



15. SETTINGS MODULE

The Settings module provides options for customizing application preferences and appearance.

It gives users a dedicated place to manage available application settings.



16. DATABASE MANAGEMENT

LifeOS uses SQLite as its database system.

SQLite is a lightweight database that stores information locally. It does not require a separate database server.

The application uses Qt SQL to communicate with the database.

The database is used to store information such as tasks and financial records, according to the implemented modules.

17. TESTING

Testing is performed to check whether the application features work as expected.

Test Case	Expected Result
Open application	Dashboard should appear
Add a task	Task should be saved
Edit a task	Task details should be updated
Delete a task	Selected task should be removed
Search tasks	Matching tasks should appear
Add financial transaction	Transaction should be recorded
Navigate between modules	Selected module should open

Testing helps identify errors and improve application reliability.

18. ADVANTAGES OF THE PROJECT

1. Multiple personal management features in one application.
2. User-friendly graphical interface.

3. Local data storage using SQLite.
4. Easy navigation between modules.
5. Helps users organize daily activities.
6. Demonstrates practical use of C++ and Qt.

19. LIMITATIONS

1. The application is currently designed as a desktop application.
2. Data is stored locally on the computer.
3. Online synchronization is not included.
4. Some modules may require further development to provide more advanced functionality.

20. FUTURE SCOPE

In the future, LifeOS can be improved by adding:

1. Cloud backup and synchronization.
2. User login and authentication.
3. Notifications and reminders.
4. Advanced financial reports.
5. More detailed productivity charts.
6. Mobile application support.
7. Data export and backup options.

21. CONCLUSION

LifeOS – Personal Digital Command Center is a C++ desktop application developed to make personal activity management easier and more organized.

The project combines different modules such as task management, calendar, habit tracking, personal finance, notes, and analytics into one application.

During the development of this project, practical knowledge of C++, Qt 6, SQLite, database connectivity, graphical user interface development, and application building was gained.

Overall, LifeOS demonstrates how software development can be used to solve everyday organizational problems through a single desktop application.

22. REFERENCES

1. C++ Programming Documentation
2. Qt 6 Official Documentation
3. SQLite Official Documentation
4. CMake Documentation
5. Class notes and programming resources